QUANTUM HARNESS · CHALLENGE 122

Three Routes to Central Charge

Clean Ising, Nishimori Disorder, and Weak Self-Dual Majorana Dynamics

Team ■■■■ · Quantum Harness Challenge #122

Abstract

This report verifies central-charge extraction in three progressively more complex critical systems using frozen numerical evidence. A clean square-lattice Ising model supplies an exact benchmark, a quenched random-bond Ising model on the Nishimori line tests disorder averaging, and a Born-correlated weak self-dual Majorana network tests Gaussian trajectory methods. The measured values are 0.498739, 0.456469, and 0.444107; their declared intervals are consistent with the corresponding targets 0.5, 0.464, and 0.447. The emphasis is not merely numerical agreement: every result is accompanied by an explanation of the estimator, finite-size ansatz, parameter choices, statistical uncertainty, systematic limitations, and independent scientific checks.

0.498739	**0.456469**	**0.444107**
Clean Ising MC	Nishimori	Weak self-dual

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SECTION 01

Executive Summary

Central charge is a compact measure of the long-distance degrees of freedom at a two-dimensional critical point. It enters the universal finite-size correction to a free energy or closely related information rate. That correction is small compared with the extensive bulk term, so a convincing numerical estimate requires more than a visually good fit. The calculation must control sampling noise, correlations, finite-width corrections, numerical stability, and the possibility that the implemented model differs subtly from the intended one.

The three studies form a deliberate ladder of difficulty. In the clean Ising model, a transfer-matrix calculation supplies a deterministic oracle and Wolff-cluster Monte Carlo supplies an independent stochastic route. The Nishimori model replaces uniform couplings with quenched random signs and therefore requires disorder averages, correlated finite-size vectors, and a Nishimori identity check. The weak self-dual network replaces equilibrium spin configurations with sequential Born-conditioned Gaussian fermion trajectories; its central-charge signal is extracted from a Shannon free-energy rate rather than the ordinary equilibrium free energy.

Table 1. Headline numerical results

Model	Estimate	SE	95% CI	Target	Runtime (s)	Gates
Clean Ising	0.498739	0.005225	[0.488719, 0.509064]	0.500	514.3	PASS
Nishimori random-bond Ising	0.456469	0.008181	[0.440064, 0.472304]	0.464	466.6	PASS
Weak self-dual Majorana network	0.444107	0.004240	[0.435749, 0.452494]	0.447	330.3	PASS

The clean row reports the Monte Carlo estimate so that all three rows carry sampling intervals. Its independent transfer-matrix estimate is 0.499424. Runtime values describe the recorded source workflows and are not directly comparable as hardware-independent costs.

INTERPRETATION

All three estimates are statistically consistent with their declared benchmarks. This means the frozen data do not resolve a discrepancy at their quoted precision; it does not mean the numerical estimates are identically equal to the target values or free of broader systematic uncertainty.

The clean estimate is c = 0.498739 with a 95% interval [0.488719, 0.509064], while the deterministic route gives 0.499424. The Nishimori estimate is c_eff = 0.456469 with interval [0.440064, 0.472304]. The weak self-dual estimate is c_eff = 0.444107 with interval [0.435749, 0.452494]. Every required gate stored with these result sets passes.

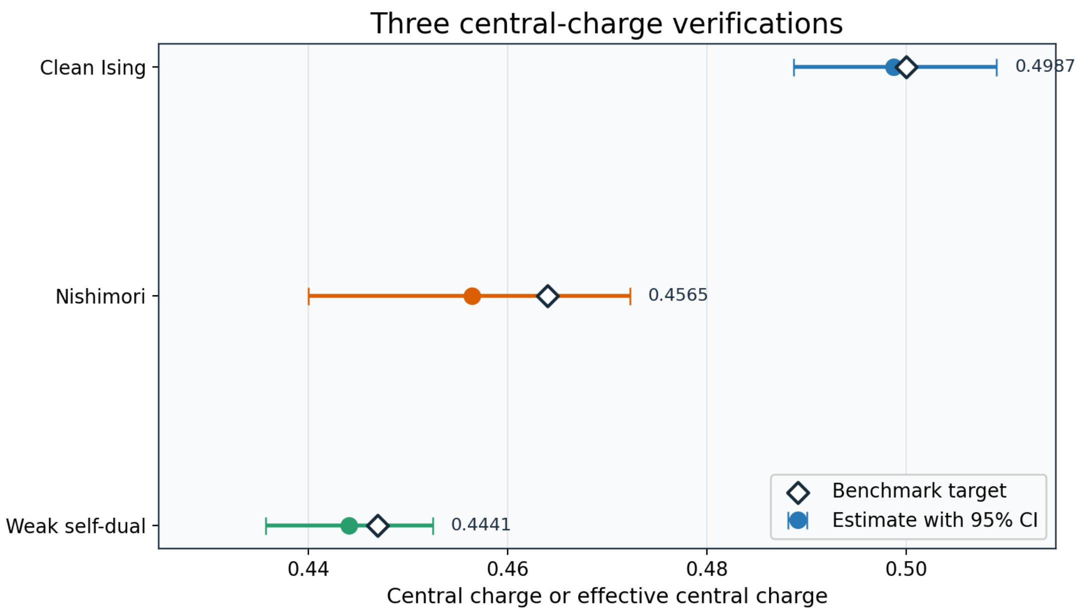

Figure 1. Cross-model result summary. Points show the measured estimates and bars show 95% confidence intervals; distinct markers identify benchmark targets rather than treating them as additional measurements. *Interpretation limit:* Interval overlap with a target establishes consistency at the reported precision, not identity of microscopic models or equality of all systematic errors.

SECTION 02

Conceptual Foundation

A continuous phase transition has fluctuations on every length scale. The correlation length diverges, local microscopic details become less important, and observables organize into scaling laws. In two spatial dimensions at equilibrium, the long-distance fixed point is often described by a conformal field theory. Conformal symmetry is much more restrictive than ordinary scale symmetry, and one of its defining numbers is the central charge c. Roughly speaking, c counts effective gapless degrees of freedom, although that phrase must be used carefully for non-unitary, disordered, or information-theoretic problems.

A finite cylinder turns central charge into a measurable correction. The leading free-energy density is nonuniversal: it depends on lattice spacing, local interactions, and normalization. The subleading Casimir term is universal because critical fluctuations wrap around the finite circumference. For a periodic cylinder of circumference L, the correction is proportional to c/L^2 in a free-energy density, or to c/L in a free energy per longitudinal step. The sign depends on the convention used for free energy, log partition function, or Shannon surprise. Each model chapter derives the convention actually fitted rather than moving signs by analogy.

```
f(L) = f_infinity - pi c/(6 L^2) + a/L^4 + ...
```
(1)

Here f(L) is an intensive critical free-energy quantity, f_infinity is the nonuniversal bulk limit, and a absorbs the leading analytic or irrelevant-operator correction. The coefficient of 1/L^2 carries c. The L^-4 term is retained because the smallest simulated widths are not asymptotic. Omitting it may reduce nominal error bars while biasing the central term.

Finite-size scaling is an inference problem with an unfavorable signal structure. The bulk contribution is order one, while the universal term shrinks as L^-2. At larger L the theoretical approximation improves, but the difference used to identify c becomes smaller. At smaller L the signal is larger, but neglected corrections are more dangerous. A fit window therefore expresses a bias-variance compromise. It must be chosen or at least tested without selecting whichever window happens to produce the expected answer.

The notation c_eff is used when disorder, non-unitarity, replicas, or an information-theoretic ensemble changes what the Casimir coefficient represents. It remains the coefficient extracted from a universal finite-size term, but it need not equal a simple count of unitary CFT fields. This is why the clean Ising benchmark is labeled c=1/2 whereas the Nishimori and weak self-dual results are labeled effective central charges.

Quenched disorder means that random couplings are drawn and then held fixed while thermal degrees of freedom equilibrate. The physical free energy involves the disorder average of log Z, not the log of the disorder-averaged Z. These operations differ by Jensen's inequality. A numerical algorithm must therefore compute or estimate log-normalization increments for each disorder history before averaging. Treating every bond as if it were re-sampled during the thermal trace would instead describe annealed disorder and a different universality class.

```
F_quenched = - E_disorder[log Z], F_annealed = -log E_disorder[Z]
```
(2)

The distinction is conceptual and computational. The Nishimori workflow samples disorder realizations, evaluates thermal transfer operations for each realization, and averages accumulated logarithms. The equality of these expressions is not assumed.

Self-duality exchanges two complementary descriptions of the same critical point. In an Ising setting it exchanges high- and low-temperature objects; in the Majorana network it exchanges electric and magnetic vortex species after a one-Majorana translation. Weak self-duality does not mean that every individual random trajectory is symmetric. It means that the correctly sampled ensemble should show the corresponding exchange symmetry. A paired density difference is therefore a useful diagnostic of model implementation and sampling.

> **WHY THREE MODELS?**
> The shared finite-size idea becomes more credible when it survives three different sources of complexity: ordinary thermal sampling, quenched disorder, and state-conditioned quantum measurement outcomes. The comparison is methodological, not a claim that the microscopic systems are interchangeable.

SECTION 03

Shared Computational Architecture

All computationally intensive sampling and state evolution are written in Rust. Rust provides predictable memory use, explicit numerical data types, parallel iterators, and a compiler that catches many indexing and ownership errors before a long run starts. Python begins only after atomic artifacts have been written. It reads records, aggregates blocks, performs regressions and bootstrap resampling, and creates charts. This division keeps exploratory statistics convenient without putting the inner simulation loop behind an interpreter.

The pseudo-random generator is Xoshiro256++, supplied by the Rust rand_xoshiro implementation. A base seed is not reused directly for every width and replica. Instead, deterministic derivation assigns a distinct stream identity from the tuple of model, width, replica or trajectory, and measurement role. Reproducibility therefore means that the same configuration and stream key recreate the same bytes, while parallel scheduling does not decide which random numbers a stream receives.

DETERMINISTIC STREAM DERIVATION

```
key = hash(base_seed, model_tag, width, replica, purpose)
rng = Xoshiro256PlusPlus::seed_from_u64(key)
for block in assigned_blocks:
    estimate = simulate_block(rng, state)
    write_atomic(stream_key, block, estimate)
```

A stream key is part of the scientific record. Atomic replacement avoids a valid-looking partial JSON file after interruption, and stable keys allow completed streams to be reused byte for byte.

Raw results are organized into blocks rather than only global averages. Blocking preserves information needed to diagnose correlation and to resample uncertainty. A single mean and variance cannot reveal whether early and late measurements disagree, whether one replica dominates, or whether widths share a common disorder realization. The precise block definition differs by model: Wolff sweeps for clean Ising, transfer rows for Nishimori disorder, and circuit layers within independent streams for the Majorana network.

Manifests record configuration values, software versions, seeds, command lines, runtimes, and cryptographic hashes. Loaders reject incompatible schema versions and mismatched configurations. This is scientific error handling rather than administrative bookkeeping. Combining a summary from one configuration with raw blocks from another can produce smooth plots and plausible numbers; schema and hash checks turn that silent failure into an explicit exception.

Table 2. Shared software responsibilities

Layer	Responsibility	Why it lives there
Rust core	State updates, transfer actions, sampling, atomic records	Performance and explicit numerical invariants
Rust tests	Small exact oracles, RNG replay, geometry and CLI contracts	Catch physics and serialization errors near implementation
Python analysis	Aggregation, fitting, bootstrap, diagnostics	Transparent numerical experimentation and mature statistics
Python plotting	Existing model figures and integrated comparisons	Consistent publication graphics
Report generator	Frozen-source validation and dual-format rendering	Separates communication from simulation

The boundary is artifact based: Python consumes immutable outputs rather than reaching into a running Rust process.

Tests are layered in the same order as the scientific argument. Unit tests establish local identities, such as a Born probability for a two-mode state. Integration tests compare complete short trajectories against dense exact evolution. Statistical gates then assess production data, and report tests verify that published values match frozen artifacts. A final visually inspected report is therefore the last link in a chain, not a substitute for testing the simulation.

SECTION 04

Clean Ising Model

The clean benchmark is the ferromagnetic square-lattice Ising model with spins s_i in {-1,+1} and nearest-neighbor energy H = -sum s_i s_j. With the coupling absorbed into K, the exact critical point is K_c = 0.5 log(1+sqrt(2)). The continuum limit is the minimal Ising CFT with central charge c=1/2. Because both K_c and c are known, this model tests the complete measurement pipeline without asking the fit to discover an unknown universality class.

$$H = -\sum_{\langle ij\rangle} s_i s_j, \quad K_c = (1/2)\log(1+\sqrt{2}) \tag{3}$$

Periodic L by M tori are evaluated with M=8L. The elongated geometry approximates a cylinder while retaining periodic boundaries and limits contamination from the longitudinal circumference.

The deterministic route applies a transfer matrix without constructing the full dense matrix. A row configuration has 2^L states. Horizontal weights are diagonal, while vertical bonds factor into local two-state operations. This factorization reduces a naive O(4^L) multiplication to O(L 2^L). Power iteration obtains the dominant eigenvalue and hence the free energy per row. Tight eigenvalue and residual tolerances make this route effectively exact for the widths used in the fit.

MATRIX-FREE TRANSFER ACTION

```
v = horizontal_weights * input
for site in 0..L:
    v = apply_two_state_vertical_factor(v, site, K)
normalize v and accumulate log_norm
repeat until eigenvalue and residual tolerances pass
```

The algorithm stores vectors of length 2^L but never a 2^L by 2^L matrix. The transfer estimate provides a deterministic oracle for the Monte Carlo and fitting conventions.

The stochastic route uses Wolff cluster updates. Near criticality, local single-spin updates suffer critical slowing down because correlated domains grow with system size. A Wolff update selects a seed spin and grows a like-spin cluster with the coupling-dependent bond probability, then flips the entire cluster. Large-scale modes move collectively, reducing autocorrelation relative to a local Metropolis chain. Stored blocks still matter because cluster updates do not make successive measurements exactly independent.

$$F(K_c) = -N \log 2 + \int_0^{K_c} \langle H\rangle_K \, dK \tag{4}$$

At K=0 the partition function is exactly 2^N. The derivative of the dimensionless free energy with respect to K is the mean energy in the chosen sign convention. Simpson integration over a 129-point coupling grid reconstructs the critical free energy. A nested 65-point grid tests quadrature convergence using measurements already present on the fine grid.

This integration strategy is used because Monte Carlo directly estimates expectation values, not an absolute partition function. Anchoring at K=0 eliminates an unknown additive constant. The price is accumulated quadrature and sampling uncertainty. The nested-grid shift is compared with the bootstrap standard error; agreement indicates that additional K points would not materially improve the present central-charge precision.

$$g(L)/L = f_{\infty} - \pi c/(6 L^2) + a/L^4 \tag{5}$$

The primary fit uses L_min=6. L=4 remains visible as a small-width diagnostic, and L_min=8 tests sensitivity after discarding more signal. The transfer and Monte Carlo routes use the same finite-size convention, so agreement simultaneously tests free-energy signs, normalization by site, and the regression coefficient that is converted into c.

CLEAN BENCHMARK RESULT

The transfer-matrix estimate is c=0.499424. The independent Wolff/integration estimate is c=0.498739 with SE=0.005225 and 95% CI [0.488719, 0.509064]. Both are consistent with the exact Ising value 0.5.

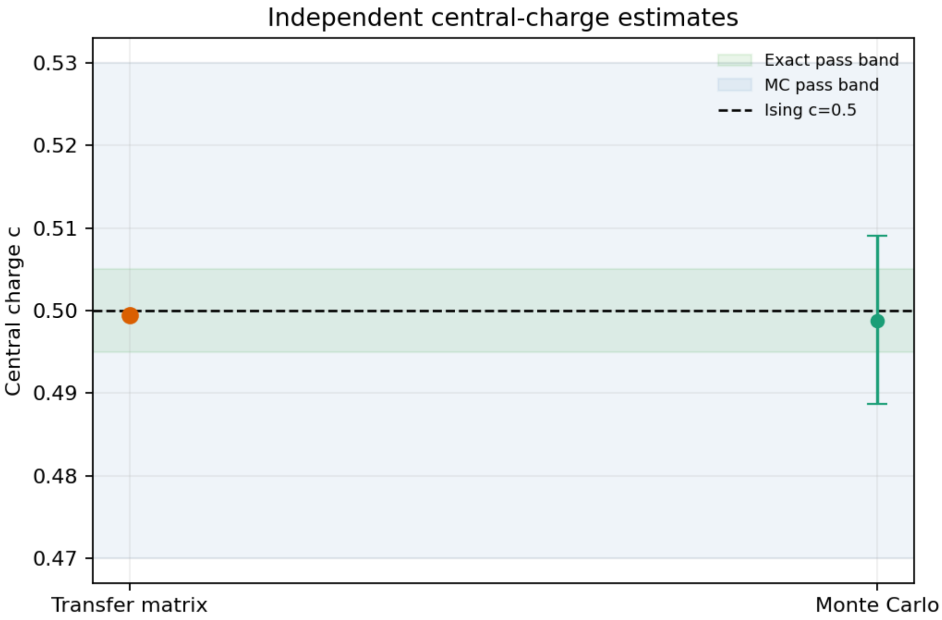

Figure 2. Independent transfer-matrix and Monte Carlo estimates are compared with c=1/2 across declared fit windows. *Interpretation limit:* Agreement does not by itself test thermalization or quadrature convergence; those appear in separate diagnostics.

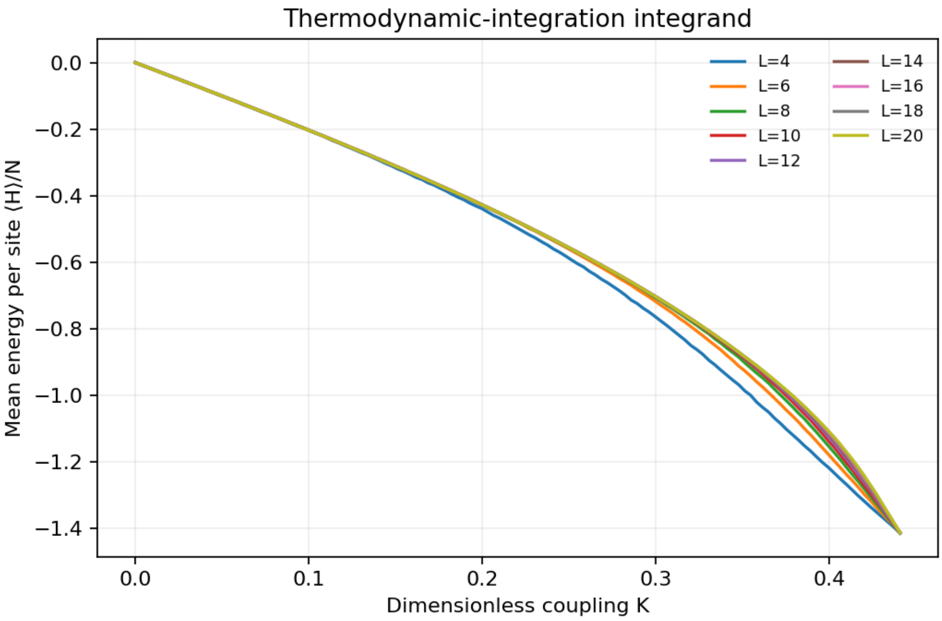

Figure 3. Measured energy density across the thermodynamic-integration grid for every width. *Interpretation limit:* Smooth curves support numerical integration but do not quantify correlation between measurements.

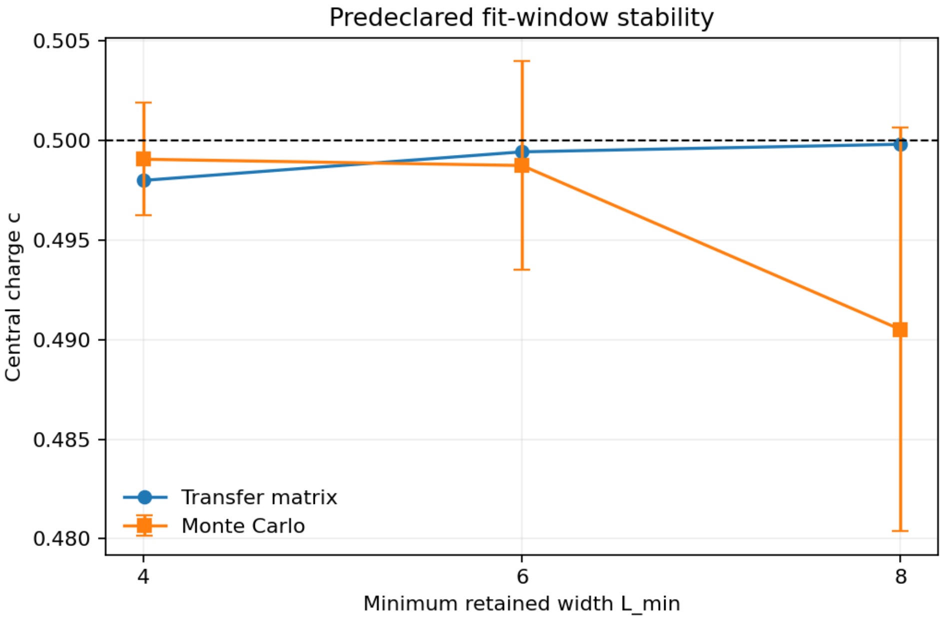

Figure 4. Central-charge estimates for L_min=4, 6, and 8 expose sensitivity to the fit window. *Interpretation limit:* The plot is diagnostic; the primary L_min=6 window is not selected after viewing it.

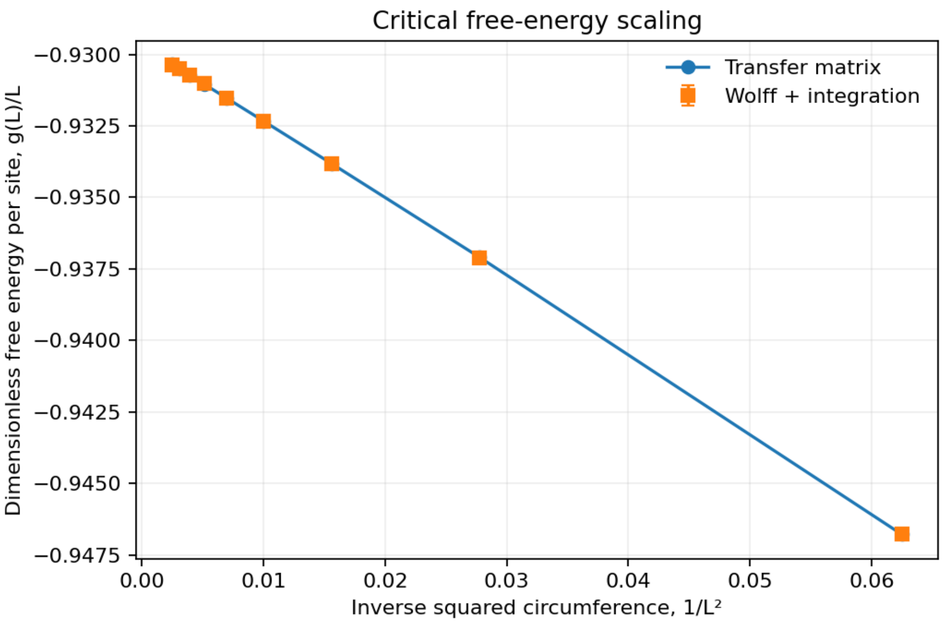

Figure 5. Critical free-energy density versus 1/L^2 for exact and Monte Carlo routes, with the fitted Casimir curvature. *Interpretation limit:* A close fit cannot rule out still higher finite-size corrections outside the simulated widths.

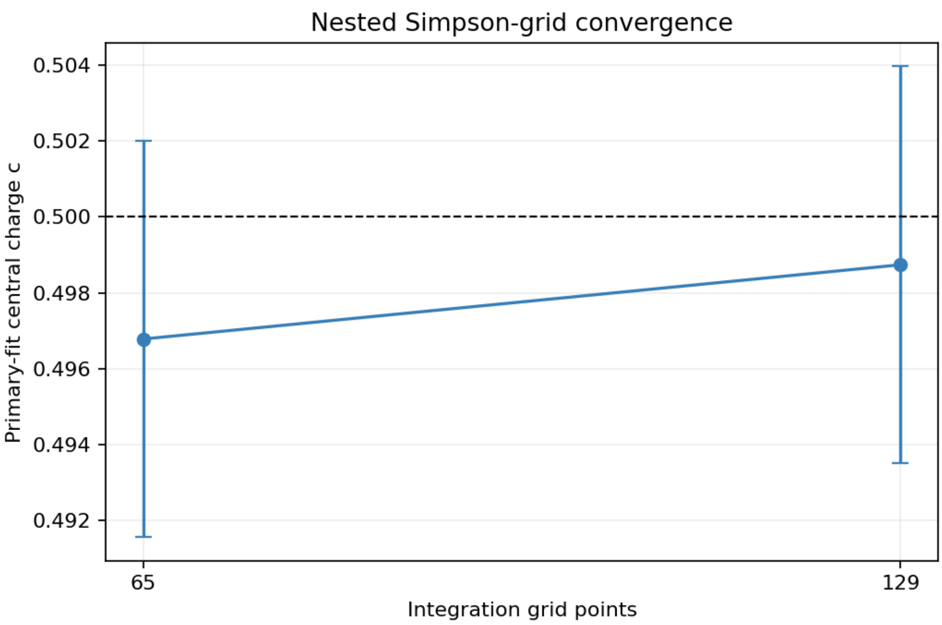

Figure 6. Nested 65- and 129-point integration grids are compared on shared nodes. *Interpretation limit:* This tests grid resolution at current sampling precision, not exact quadrature error.

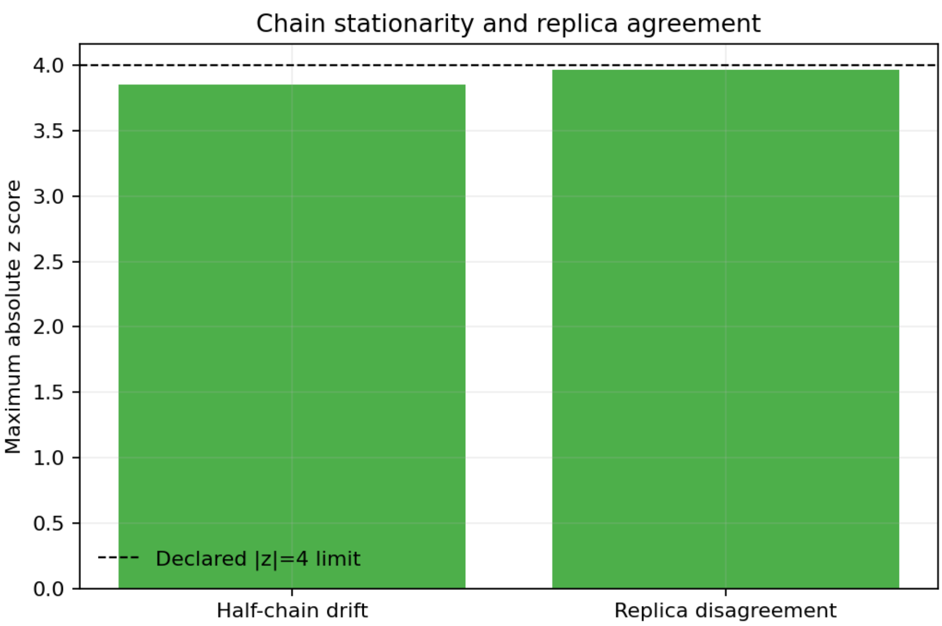

Figure 7. Half-chain drift and pairwise replica discrepancies are compared with predeclared thresholds. *Interpretation limit:* Passing finite diagnostics cannot prove a chain has sampled every exponentially rare configuration.

Table 3. Clean Ising production parameters

Symbol	Value	Meaning	Sensitivity
K_c	0.4406867935097714	Exact square-lattice critical coupling	Fixes the evaluation point
L	4, 6, 8, 10, 12, 14, 16, 18, 20	Periodic cylinder circumference	Controls finite-size bias
M/L	8	Torus aspect ratio	Suppresses longitudinal finite-size effects
N_K	129	Thermodynamic-integration grid points	Controls quadrature error
R	4	Independent Monte Carlo replicas	Supports between-chain diagnostics
N_therm	200	Discarded Wolff sweeps	Controls initialization bias
N_meas	12800	Measured Wolff sweeps per grid point	Controls statistical precision
N_block	320	Sweeps per stored block	Reduces autocorrelation
RNG	Xoshiro256++	Deterministic pseudo-random generator	Enables reproducible independent streams

Values come from the frozen production manifest or the recorded algorithm contract. Sensitivity describes the main scientific role rather than a dimensionless derivative.

Parameter K_c is set to 0.4406867935097714. It represents exact square-lattice critical coupling. Its principal role is that it fixes the evaluation point. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter L is set to 4, 6, 8, 10, 12, 14, 16, 18, 20. It represents periodic cylinder circumference. Its principal role is that it controls finite-size bias. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter M/L is set to 8. It represents torus aspect ratio. Its principal role is that it suppresses longitudinal finite-size effects. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_K is set to 129. It represents thermodynamic-integration grid points. Its principal role is that it controls quadrature error. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter R is set to 4. It represents independent monte carlo replicas. Its principal role is that it supports between-chain diagnostics. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_therm is set to 200. It represents discarded wolff sweeps. Its principal role is that it controls initialization bias. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_meas is set to 12800. It represents measured wolff sweeps per grid point. Its principal role is that it controls statistical precision. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_block is set to 320. It represents sweeps per stored block. Its principal role is that it reduces autocorrelation. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter RNG is set to Xoshiro256++. It represents deterministic pseudo-random generator. Its principal role is that it enables reproducible independent streams. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Table 4. Clean Ising validation gates

Gate	Criterion	Observed value	Required	Status
exact_accuracy	transfer-matrix estimate agrees with c=1/2	True	yes	PASS
mc_accuracy	Monte Carlo estimate agrees with c=1/2	True	yes	PASS
mc_interval	Monte Carlo 95% CI contains c=1/2	True	yes	PASS
integration	nested thermodynamic-integration grids agree	True	yes	PASS
exact_window	exact fit is stable to L_min	True	yes	PASS
mc_window	Monte Carlo fit is stable to L_min	True	yes	PASS
thermalization	half-chain drift is below threshold	True	yes	PASS
replicas	replica disagreement is below threshold	True	yes	PASS
runtime	runtime is below the declared limit	True	yes	PASS

A required failure stops the production analysis. Gate counts should not be used to rank models because the checks constrain different failure modes.

The main statistical limitation is the thermodynamic-integration noise propagated through the finite-size slope. The exact route is much more precise, but it is not a replacement for Monte Carlo: the challenge is to verify a simulation architecture that can later operate when no transfer oracle is available. Agreement between methods is strongest when each route retains independent code and numerical assumptions.

SECTION 05

Nishimori Random-Bond Ising Model

The Nishimori model assigns a sign tau_ij to every nearest-neighbor bond. A positive sign is ferromagnetic and a negative sign is antiferromagnetic. The negative-bond probability p creates frustration: no spin assignment can satisfy every bond around a plaquette with an odd product of negative signs. Disorder is quenched, so the thermal partition function is evaluated for a fixed bond history before logarithms are averaged across histories.

```
P(tau_ij=-1)=p, K_N=(1/2) log((1-p)/p)
```
(6)

The Nishimori line ties the coupling K_N to the disorder probability. At the chosen multicritical parameters, p=0.1092212 and K_N=1.049360476302568. This relation produces exact gauge identities that are valuable implementation checks.

The target in this report is the ordinary quenched effective central charge near 0.464. It must not be confused with a different Born-rule or higher-replica quantity near 0.522. Both numbers can arise in related literature, but they correspond to different ensemble weights. The implemented calculation averages log Z for ordinary quenched disorder; its data and gates were therefore designed to verify 0.464 rather than retrofit an estimate to 0.522.

For each random transfer row, Rust applies the horizontal and vertical bond factors to a vector over 2^L boundary spin states. L1 normalization after every row prevents overflow. The accumulated logarithms of the normalization factors converge to a leading Lyapunov exponent of the random transfer product. Dividing by row count and width produces the quenched log-partition density phi_L.

QUENCHED TRANSFER-PRODUCT ESTIMATOR

```
for replica in disorder_replicas:
    v = positive_initial_vector()
    for row in burn_in + measured_rows:
        tau = sample_bonds(xoshiro256pp)
        v = apply_random_transfer(v, tau, K_N)
        scale = l1_norm(v)
        v /= scale
        if measured: block_log_norm += log(scale)
average block_log_norm/(rows*L) after each disorder history
```

Normalization is not an arbitrary numerical trick: each removed log scale is precisely the incremental free-energy information needed by the Lyapunov estimator.

A maximum-width random row is sliced into nested prefixes for all smaller widths. This common-disorder construction sharply reduces noise in the difference across L, which determines the Casimir slope. It also creates cross-width covariance. The bootstrap must therefore resample an entire width vector together. Resampling each width independently would destroy the designed covariance and misstate the uncertainty of c_eff.

```
phi_L = phi_infinity + pi c_eff/(6 L^2) + a/L^4
```
(7)

The sign is positive because phi denotes a log-partition density rather than the conventional negative free energy. The fit includes L=4 through 14 and an L^-4 correction. A frozen L_min=6 refit tests whether the smallest width drives the result.

Eight independent disorder replicas each contain 2,097,152 measured rows after 4,096 burn-in rows. Blocks contain 16,384 rows. Long blocks preserve short-range serial dependence within a transfer product, while multiple replicas reveal whether one disorder stream dominates. A paired bootstrap resamples replica-block units while keeping their vector of widths intact.

```
d phi/dK |_(K_N) = 2 tanh(K_N)
```
(8)

The Nishimori energy identity is evaluated through a centered common-disorder finite difference with delta K=10^-4. Reusing exactly the same bond rows on both sides greatly reduces variance. Agreement checks the coupling convention, bond signs, transfer normalization, and the configured Nishimori relation at once.

NISHIMORI RESULT

The ordinary quenched estimate is c_eff=0.456469, SE=0.008181, and 95% CI [0.440064, 0.472304]. The target 0.464 lies inside the interval, and all nine required validation gates pass.

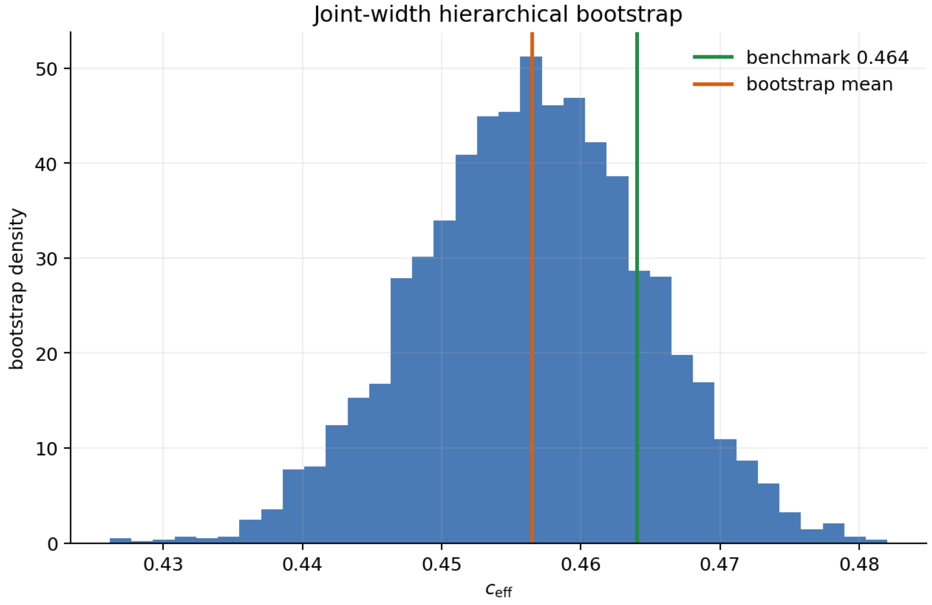

Figure 8. Hierarchical paired-bootstrap distribution of the Nishimori effective central charge. *Interpretation limit:* The interval assumes the chosen replica-block units adequately represent disorder fluctuations.

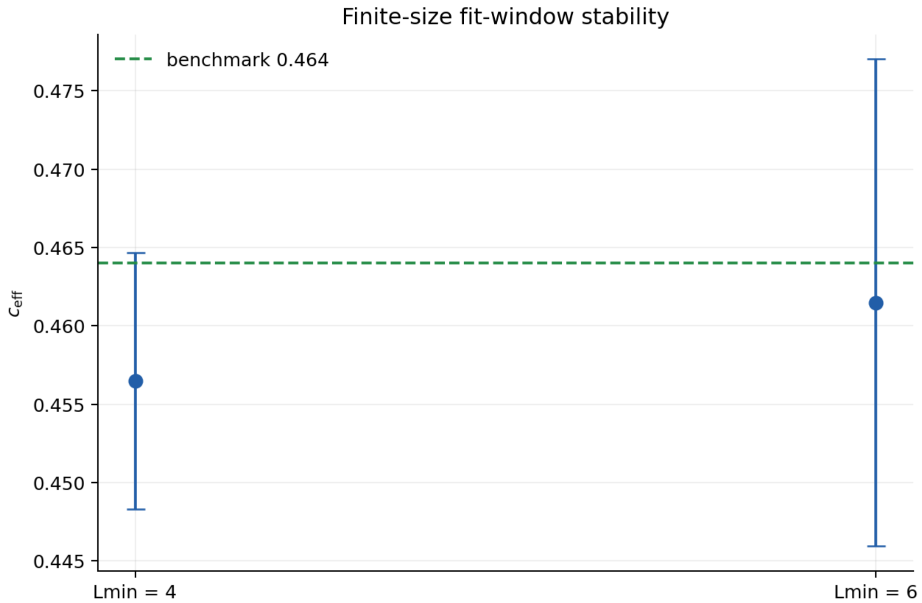

Figure 9. Primary L_min=4 and diagnostic L_min=6 fits are compared using paired resampling. *Interpretation limit:* Two windows do not exhaust all possible irrelevant-operator corrections.

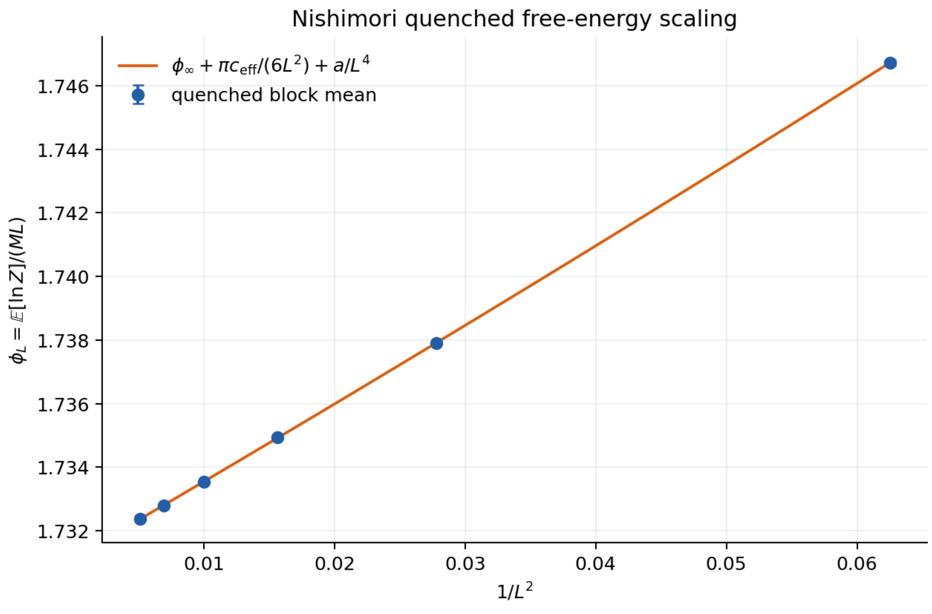

Figure 10. Quenched log-partition density versus 1/L^2 with the L^-4 corrected finite-size fit. *Interpretation limit:* Small residuals alone cannot validate the quenched ensemble or bond generator.

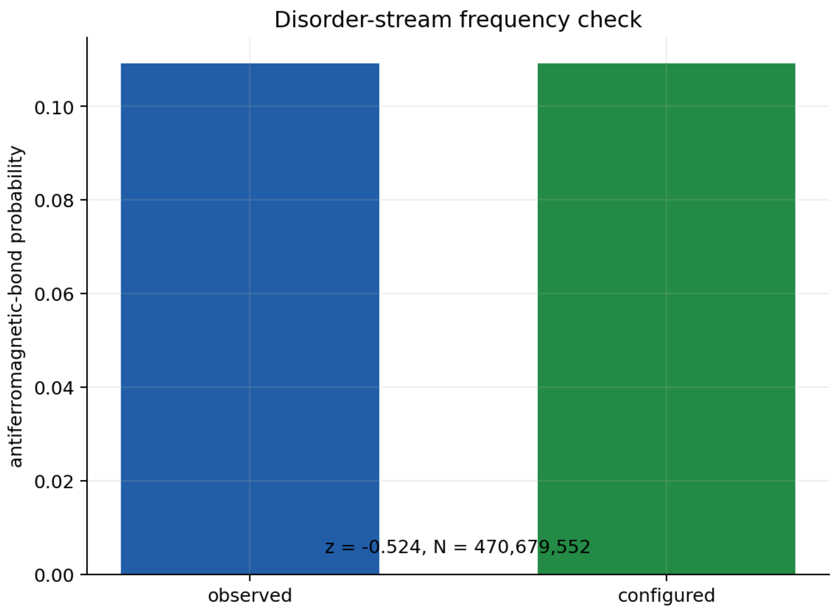

Figure 11. Observed negative-bond frequency is compared with configured p using a z score. *Interpretation limit:* The marginal frequency does not test every spatial or temporal correlation of the RNG stream.

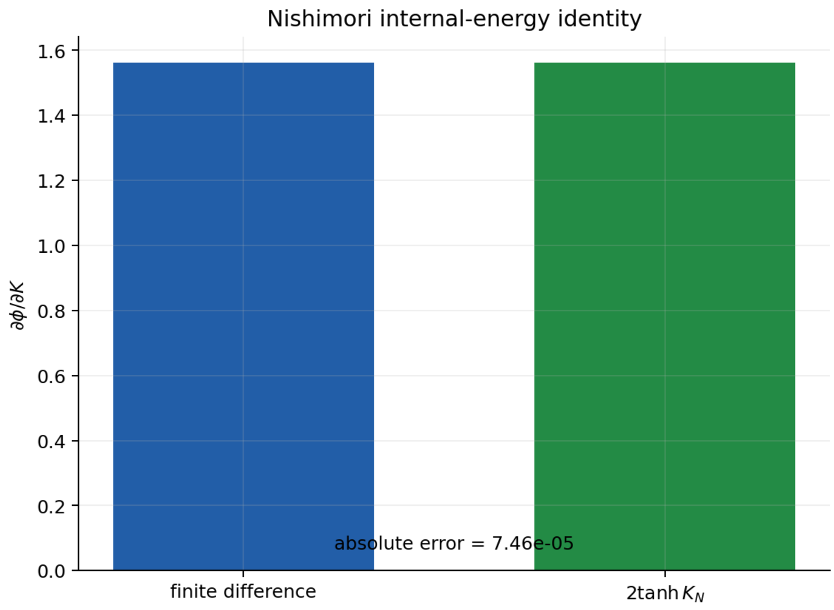

Figure 12. A common-disorder centered derivative is compared with the exact Nishimori identity. *Interpretation limit:* One identity strongly tests conventions but cannot identify every possible transfer-kernel defect.

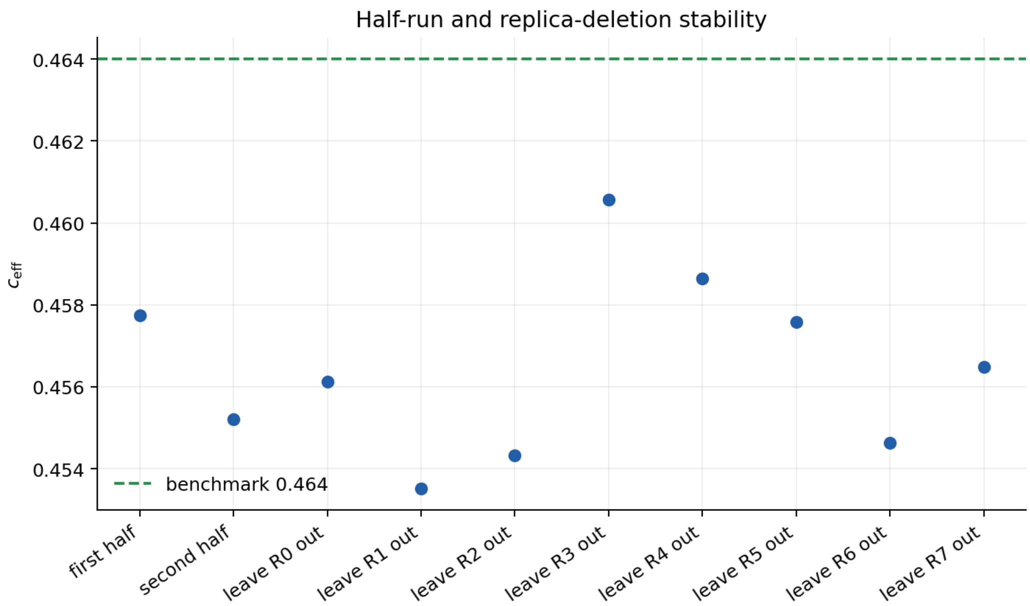

Figure 13. First/second-half and leave-one-replica-out estimates reveal time or replica dominance. *Interpretation limit:* These diagnostics have finite power against rare-disorder tails.

Table 5. Nishimori random-bond Ising production parameters

Symbol	Value	Meaning	Sensitivity
p	0.1092212	Probability of a negative bond	Defines the disorder distribution
K_N	1.049360476302568	Nishimori-line coupling	Locks thermal and disorder weights
L	4, 6, 8, 10, 12, 14	Transfer-strip widths	Controls finite-size bias
R	8	Independent disorder replicas	Controls disorder-sampling error
N_burn	4,096	Discarded transfer rows	Suppresses initial-vector bias
N_meas	2,097,152	Measured rows per replica	Controls free-energy precision
N_block	16,384	Rows per bootstrap block	Preserves serial dependence
delta K	1.0e-04	Centered finite-difference step	Controls energy-identity truncation
RNG	Xoshiro256++	Deterministic disorder generator	Preserves reproducible common-disorder streams

Values come from the frozen production manifest or the recorded algorithm contract. Sensitivity describes the main scientific role rather than a dimensionless derivative.

Parameter p is set to 0.1092212. It represents probability of a negative bond. Its principal role is that it defines the disorder distribution. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter K_N is set to 1.049360476302568. It represents nishimori-line coupling. Its principal role is that it locks thermal and disorder weights. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter L is set to 4, 6, 8, 10, 12, 14. It represents transfer-strip widths. Its principal role is that it controls finite-size bias. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter R is set to 8. It represents independent disorder replicas. Its principal role is that it controls disorder-sampling error. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_burn is set to 4,096. It represents discarded transfer rows. Its principal role is that it suppresses initial-vector bias. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_meas is set to 2,097,152. It represents measured rows per replica. Its principal role is that it controls free-energy precision. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_block is set to 16,384. It represents rows per bootstrap block. Its principal role is that it preserves serial dependence. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter delta K is set to 1.0e-04. It represents centered finite-difference step. Its principal role is that it controls energy-identity truncation. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter RNG is set to Xoshiro256++. It represents deterministic disorder generator. Its principal role is that it preserves reproducible common-disorder streams. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Table 6. Nishimori random-bond Ising validation gates

Gate	Criterion	Observed value	Required	Status
target_agreement	|c_eff - 0.464| <= 0.020	0.0075306	yes	PASS
target_in_confidence_interval	95% CI contains 0.464	[0.440064, 0.472304]	yes	PASS
standard_error	SE(c_eff) <= 0.010	0.00818052	yes	PASS
fit_window_agreement	paired-bootstrap 95% CI for c(Lmin=4)-c(Lmin=6) contains zero	[-0.032232, 0.0211545]	yes	PASS
half_run_stability	first/second-half discrepancy < 4 sigma	0.170997	yes	PASS
replica_stability	maximum leave-one-replica-out discrepancy < 4 sigma	0.501081	yes	PASS
nishimori_energy_identity	|d phi/dK - 2 tanh(K_N)| <= 0.005	7.46036e-05	yes	PASS
negative_bond_frequency	|bond-frequency z| < 4	-0.524227	yes	PASS

Gate	Criterion	Observed value	Required	Status
runtime	end-to-end runtime < 600 s	466.618	yes	PASS

A required failure stops the production analysis. Gate counts should not be used to rank models because the checks constrain different failure modes.

The dominant uncertainty is disorder sampling rather than thermal spin sampling, because the transfer operation sums thermal boundary states deterministically for each random row. Finite-width corrections remain a separate systematic concern. The paired fit-window interval contains zero, so the present data do not resolve a significant L_min shift, but that diagnostic does not prove that every higher-order correction is negligible.

SECTION 06

Weak Self-Dual Majorana Network

The weak self-dual model is a monitored Majorana network rather than an equilibrium spin system. Its state is Gaussian, so it can be represented by a real antisymmetric covariance matrix Gamma instead of an exponentially large many-body wavefunction. A measurement layer alternates onsite and bond Majorana bilinears. At theta=pi/4 and equal weak couplings, a one-Majorana translation exchanges the electric and magnetic descriptions.

Gaussian covariance methods make the production widths feasible, but they must preserve physical constraints. For a pure state, Gamma is antisymmetric and Gamma squared is -I up to floating-point error. Weak measurement updates are fractional-linear transformations of Gamma. Periodic stabilization projects numerical drift back toward the pure Gaussian manifold, and the maximum invariant error is recorded rather than silently corrected.

```
P(s|Gamma) = [1 + s tanh(beta) <i gamma_a gamma_b>_Gamma]/2
```
(9)

The binary outcome s is sampled from the current state, so outcomes are Born-correlated in space and time. Replacing them with independent fair signs would define a different disorder ensemble. With beta=asinh(1)=0.881373587019543, each probability depends on the measured bilinear expectation encoded by Gamma.

A direct Shannon estimator would accumulate the realized surprise -log P(s|Gamma). The implementation uses a Rao-Blackwellized alternative: before drawing s, it records the conditional binary entropy of that measurement. Averaging the conditional expectation removes the extra coin-flip variance while leaving the ensemble mean unchanged. This variance reduction made the target precision practical without changing the Born trajectory used to update the state.

```
H_2(q) = -q log q - (1-q) log(1-q)
```
(10)

Here q is the conditional probability of one outcome. The entropy is summed over all measurements and normalized per measurement row. One circuit period contains two rows, onsite and bond. Forgetting this factor of two would rescale the extensive rate and its Casimir coefficient.

BORN-CORRELATED GAUSSIAN TRAJECTORY

```
Gamma = vacuum_covariance(L)
for layer in burn_in + measured_layers:
    for measurement_row in [onsite, bond]:
        for (a,b) in row_pairs:
            q = born_probability(Gamma, a, b, beta)
            if measured: entropy_sum += binary_entropy(q)
            s = sample_bernoulli(q, xoshiro256pp)
            Gamma = weak_gaussian_update(Gamma, a, b, s, beta)
    periodically_stabilize_and_check(Gamma)
```

The entropy record is Rao-Blackwellized, but the state update still uses the sampled outcome. Consequently the method reduces estimator variance without replacing the physical trajectory distribution.

```
gamma_1(L) = f_infinity L - pi c_eff/(6 L) + a/L^3
```
(11)

gamma_1 is an entropy or Shannon free-energy rate per longitudinal layer. It is extensive in circumference, so the universal term scales as 1/L instead of 1/L^2. Dividing the equation by L recovers the same structural Casimir scaling used in the other chapters.

The final refinement uses even widths 6 through 32, 128 independent streams per width, 20L burn-in layers, and 200L measured layers. Blocks contain 5L layers. Even widths preserve the network geometry and sector conventions. Increasing both length and stream count addresses two different uncertainties: longer streams stabilize time averages, while more streams improve independent-trajectory sampling and bootstrap reliability.

Self-duality is checked using electric and magnetic vortex densities computed from spacetime sign tiles. A one-edge defect creates adjacent vortices of the two species, providing a small exact geometry test. In production, the paired electric-minus-magnetic difference is divided by its standard error. The observed z score is -1.460, inside the declared two-sided 95% threshold.

WEAK SELF-DUAL RESULT

The estimate is c_eff=0.444107, SE=0.004240, and 95% CI [0.435749, 0.452494]. The target 0.447 is inside the interval, whose half-width is about 0.00837. All ten required gates pass.

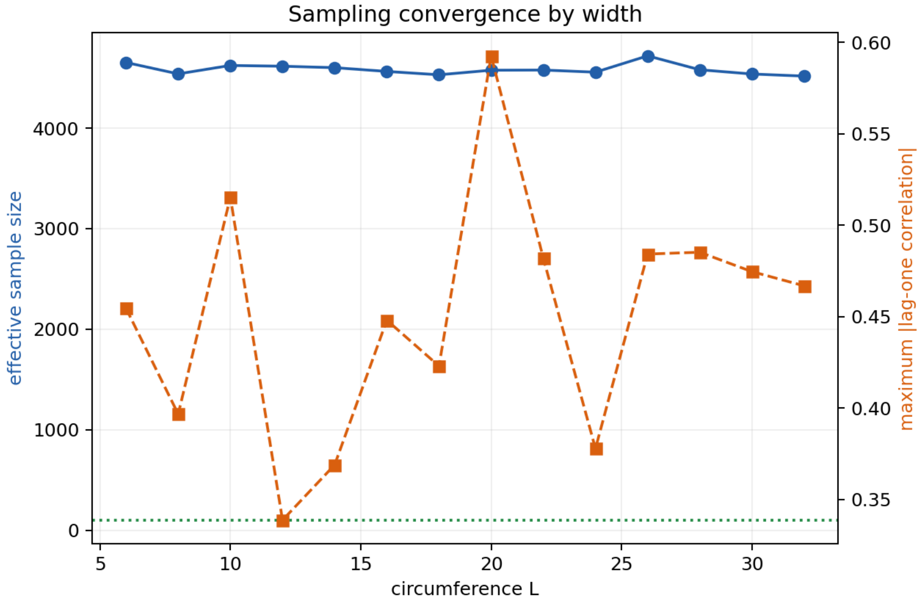

Figure 14. Effective sample size and lag-one correlation summarize trajectory convergence by width. *Interpretation limit:* ESS diagnoses sampling precision, not finite-size model bias.

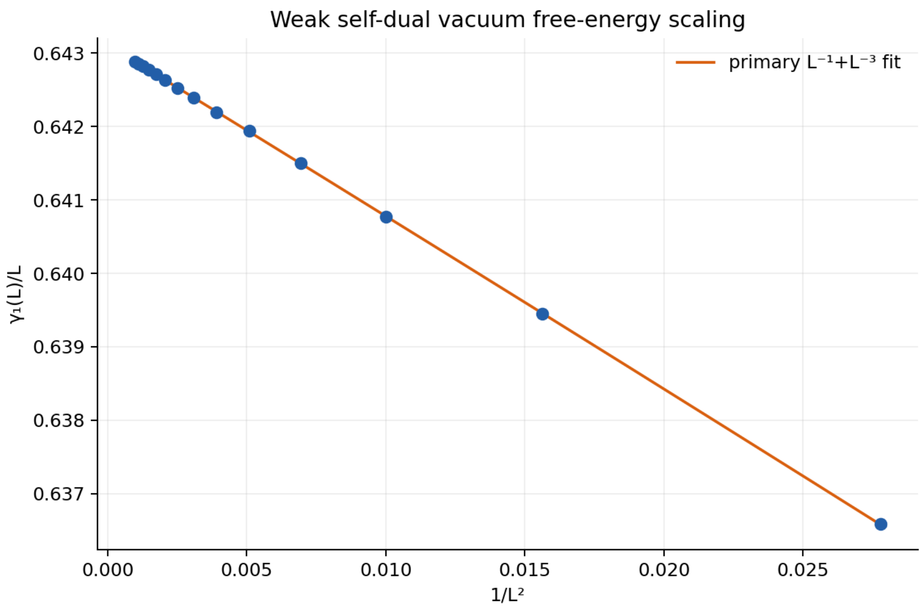

Figure 15. The Shannon free-energy rate gamma_1(L) is fitted to an extensive term plus 1/L Casimir and 1/L^3 correction. *Interpretation limit:* The visual scale is dominated by the bulk term; residual and fit-variant plots are needed to judge the Casimir coefficient.

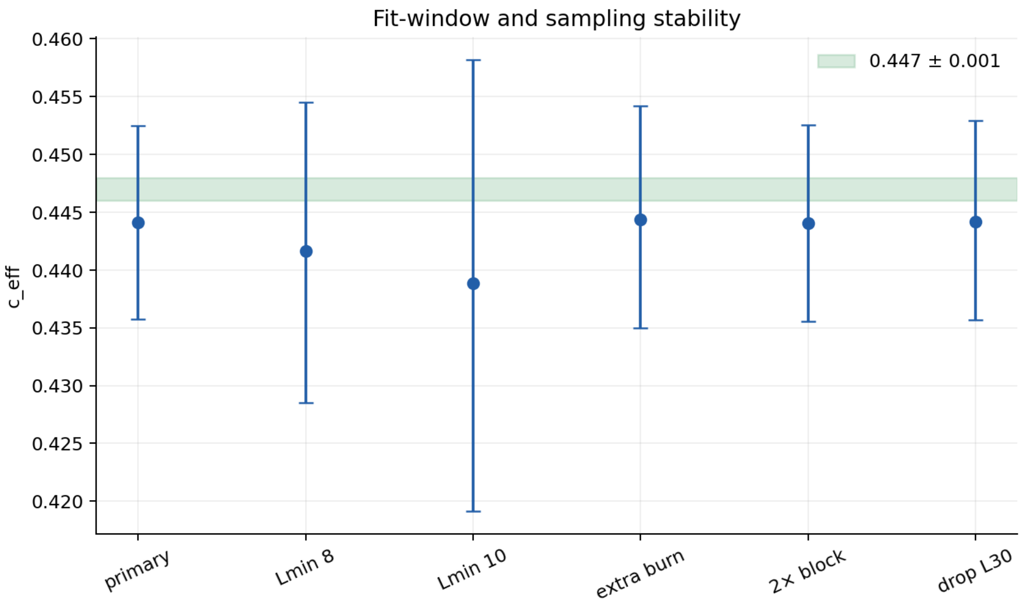

Figure 16. Paired fit variants change minimum width, burn-in, block length, and included widths. *Interpretation limit:* The tested variants bound declared sensitivities but not every conceivable correction model.

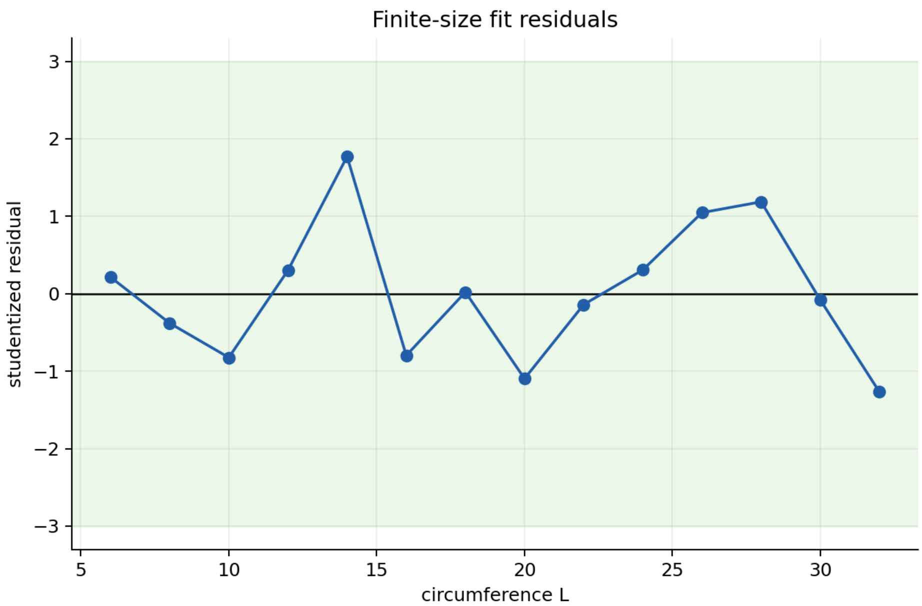

Figure 17. Studentized residuals and their trend against inverse width expose unresolved structure. *Interpretation limit:* Unstructured residuals support the fit but do not prove the asymptotic expansion is unique.

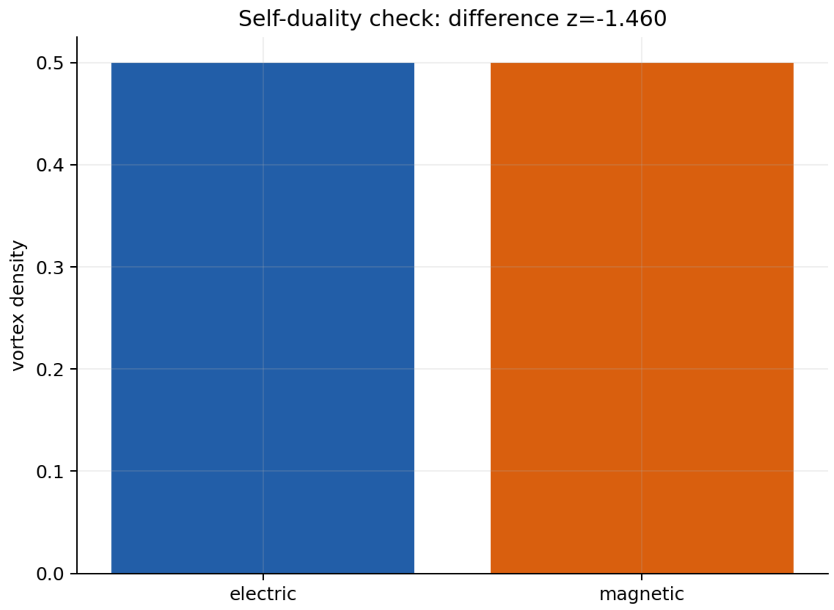

Figure 18. Electric and magnetic vortex densities are compared through their paired difference. *Interpretation limit:* Ensemble-level self-duality does not require equality on each individual trajectory.

Table 7. Weak self-dual Majorana network production parameters

Symbol	Value	Meaning	Sensitivity
theta	pi/4	Self-dual circuit angle	Fixes the isotropic weak self-dual point
beta	0.881373587019543	Weak-measurement coupling	Sets Born update strength
L	6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32	Majorana cylinder circumference	Controls finite-size bias
R	128	Independent trajectory streams per width	Controls trajectory variance
N_burn/L	20	Burn-in layers per width	Suppresses boundary transients
N_meas/L	200	Measured layers per width	Controls entropy-rate precision
N_block/L	5	Layers per stored block	Supports autocorrelation estimates
N_stab	4	Layers between covariance stabilization	Controls invariant drift
epsilon_Gamma	1e-09	Gaussian invariant tolerance	Rejects numerically invalid trajectories
RNG	Xoshiro256++	Deterministic Born sampler	Enables replayable trajectory streams

Values come from the frozen production manifest or the recorded algorithm contract. Sensitivity describes the main scientific role rather than a dimensionless derivative.

Parameter theta is set to pi/4. It represents self-dual circuit angle. Its principal role is that it fixes the isotropic weak self-dual point. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter beta is set to 0.881373587019543. It represents weak-measurement coupling. Its principal role is that it sets born update strength. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter L is set to 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32. It represents majorana cylinder circumference. Its principal role is that it controls finite-size bias. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter R is set to 128. It represents independent trajectory streams per width. Its principal role is that it controls trajectory variance. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_burn/L is set to 20. It represents burn-in layers per width. Its principal role is that it suppresses boundary transients. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_meas/L is set to 200. It represents measured layers per width. Its principal role is that it controls entropy-rate precision. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_block/L is set to 5. It represents layers per stored block. Its principal role is that it supports autocorrelation estimates. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter N_stab is set to 4. It represents layers between covariance stabilization. Its principal role is that it controls invariant drift. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter epsilon_Gamma is set to 1e-09. It represents gaussian invariant tolerance. Its principal role is that it rejects numerically invalid trajectories. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Parameter RNG is set to Xoshiro256++. It represents deterministic born sampler. Its principal role is that it enables replayable trajectory streams. This value is not a post-fit adjustment: it belongs to the frozen simulation contract. Changing it can alter computational cost, sampling correlation, finite-size bias, or even the physical ensemble, depending on the parameter. A valid sensitivity study would create a separately identified run and compare paired observables where possible rather than editing the record after inspecting the central-charge estimate.

Table 8. Weak self-dual Majorana network validation gates

Gate	Criterion	Observed value	Required	Status
exact_oracles	all exact oracle tolerances pass	True	yes	PASS
gaussian_invariants	maximum invariant error <= 1e-09	2.53797e-13	yes	PASS
self_duality	|electric-minus-magnetic z| <= 1.96	-1.45978	yes	PASS
effective_sample_size	minimum ESS >= 100	4517.15	yes	PASS
target_interval	95% CI contains 0.447	[0.435749, 0.452494]	yes	PASS
precision	95% half-width <= 0.01	0.00837276	yes	PASS
fit_stability	maximum paired fit shift < 2 sigma	0.593755	yes	PASS
systematic_spread	required fit centers span <= 0.01	0.00552828	yes	PASS
residuals	maximum |studentized residual| < 3	1.77161	yes	PASS
residual_trend	|corr(residual,1/L)| < 0.8	-0.055771	yes	PASS

A required failure stops the production analysis. Gate counts should not be used to rank models because the checks constrain different failure modes.

The strongest remaining systematic is finite-size model choice. The primary L^-3 correction, alternative minimum widths, a dropped large width, doubled blocks, and extra burn-in produce a systematic center spread of about 0.00553. The maximum paired shift is only 0.594 standard errors. These tests support stability at the claimed resolution while remaining honest about the difference between sampling error and fit model uncertainty.

SECTION 07

Cross-Model Comparison

The most useful comparison is a map from a universal idea to three model-specific realizations. In every case, a long cylinder or strip has an extensive bulk contribution plus a circumference-dependent Casimir term. The central charge is inferred from the coefficient of that small term. What changes is the object whose logarithm or information rate is accumulated and the probability measure under which it is averaged.

Table 9. Estimator and validation comparison

Model	Finite-size observable	Sampled randomness	Strong oracle	Target z
Clean Ising	Thermodynamically integrated free-energy density	Thermal Wolff clusters	Exact transfer matrix and c=1/2	-0.24
Nishimori random-bond Ising	Quenched transfer Lyapunov density	Quenched bond rows	Nishimori energy identity	-0.92
Weak self-dual Majorana network	Born Shannon free-energy rate	State-conditioned Born outcomes	Dense Gaussian/Born trajectory agreement	-0.68

Target z is (estimate-target)/SE and is shown only as a normalized summary. The fitted datasets, correction models, and error structures differ, so equal z values would not imply equal evidential strength.

In clean Ising, Monte Carlo samples thermal spin configurations and thermodynamic integration reconstructs an absolute free energy from an exact anchor. In Nishimori Ising, the transfer step sums thermal boundary spins while Monte Carlo samples quenched bonds; log normalization factors give a disorder-averaged Lyapunov rate. In the weak self-dual network, Monte Carlo samples state-dependent Born outcomes and covariance updates carry memory from one measurement to the next.

The error bars are therefore not interchangeable. The clean result propagates correlated energy estimates across a coupling grid. The Nishimori result must preserve covariance across widths created by common disorder. The weak self-dual result must capture autocorrelation within streams and heterogeneity across independent trajectories. Bootstrap resampling is used in all three analyses, but the resampling unit follows the data-generating process rather than a universal recipe.

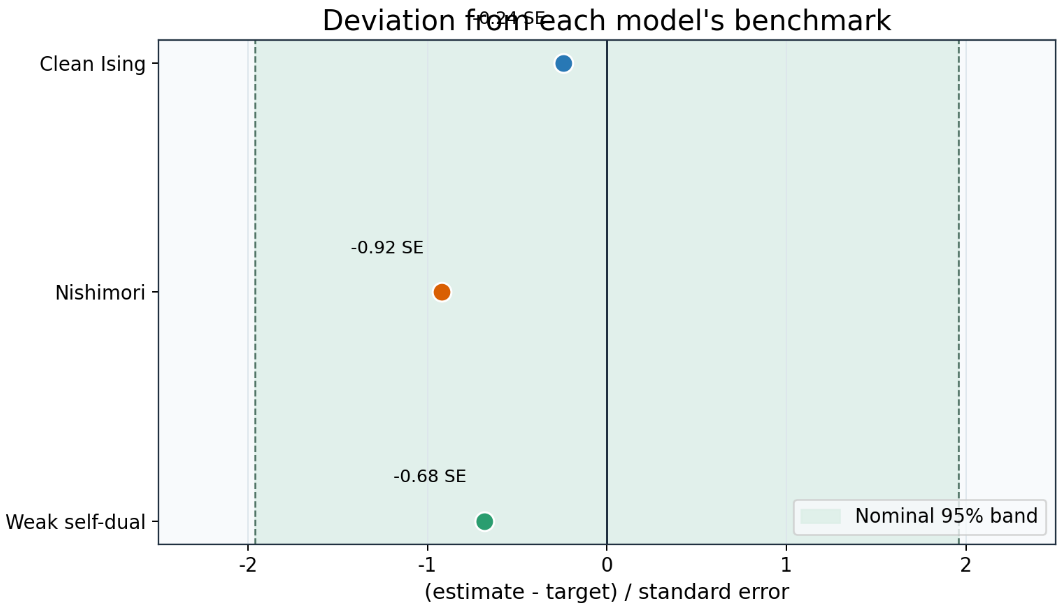

Figure 19. Each point is the estimate-minus-target difference divided by its reported standard error. The reference band makes statistical consistency easy to read without hiding the absolute intervals. *Interpretation limit:* A small normalized deviation cannot diagnose shared model bias, incorrect ensemble choice, or underestimated systematic uncertainty.

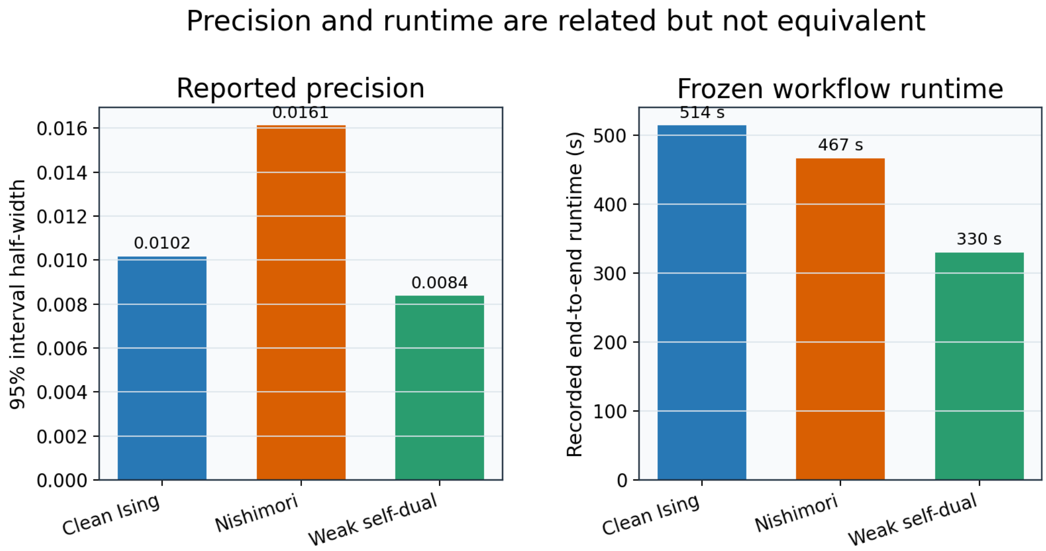

Figure 20. Precision and runtime are displayed on separate axes and panels to avoid a dual-axis visual correlation. The chart describes these frozen runs. *Interpretation limit:* Runtime depends on algorithms, widths, sample budgets, compilation, and hardware; it is not a universal complexity benchmark.

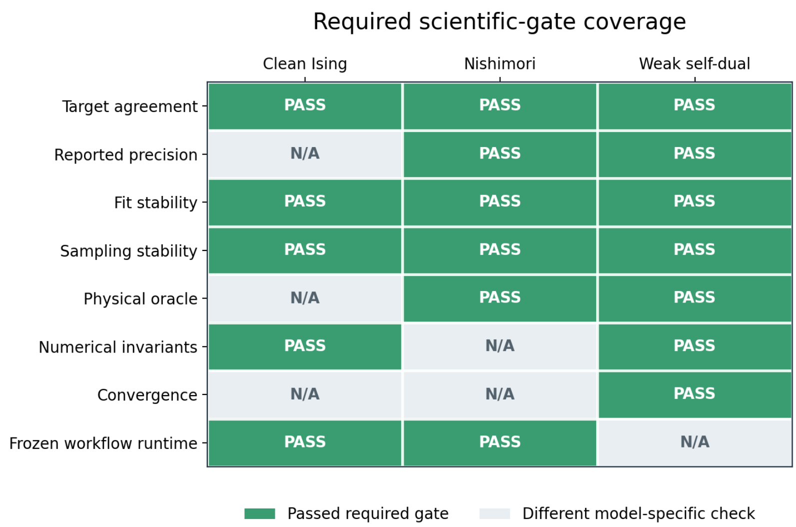

Figure 21. Green cells represent passed required checks; neutral cells indicate that a model uses a different oracle rather than lacking validation. *Interpretation limit:* Counting gates does not rank scientific quality because one strong exact oracle may constrain more failure modes than several narrow diagnostics.

SHARED IMPLEMENTATION PRINCIPLE

Estimate the small universal term only after validating the much larger bulk calculation. Exact identities, symmetry checks, replay tests, and invariant bounds are designed to fail when the model or normalization is wrong, even if a flexible finite-size fit could still land near the literature value.

The progression also explains why variance reduction changes form. Wolff updates attack critical slowing down in configuration space. Common disorder attacks variance in finite-size differences. Rao-Blackwellization removes conditional measurement noise. Each method exploits structure in its probability model; applying it outside that structure could bias the answer.

SECTION 08

Error and Sensitivity Analysis

Uncertainty is separated into sampling, numerical, finite-size, and model components. A single confidence interval usually quantifies only part of this hierarchy. Reporting one number without identifying its resampling unit can be actively misleading: treating correlated blocks as independent shrinks error bars, while treating common-width disorder as independent throws away useful covariance and can inflate them.

Autocorrelation reduces the amount of independent information in a time series. Blocking replaces many closely related measurements with fewer block summaries whose between-block variation is more informative. The weak self-dual analysis additionally estimates effective sample size. Its minimum ESS exceeds 4,500, comfortably above the declared gate of 100. That large margin supports the bootstrap but does not remove finite-size systematics.

The clean workflow contains a distinct quadrature error because free energy is reconstructed by integrating mean energy over K. Simpson's rule has high deterministic order for smooth functions, but the integrand is itself noisy. The nested-grid difference is therefore interpreted relative to the bootstrap variation, not as an exact truncation error. Measurements on shared K nodes are kept aligned during resampling.

The Nishimori workflow's major challenge is quenched heterogeneity. A rare disorder region can affect a long product of transfer operators. First-versus-second-half and leave-one-replica-out fits test whether the result depends on time position or one stream. The negative-bond frequency checks the input distribution, and the energy identity checks a derivative of the output. Passing both constrains different ends of the computation.

The weak self-dual workflow adds floating-point manifold error. Covariance updates involve matrix inverses or equivalent low-rank formulas whose roundoff can accumulate. Antisymmetry, purity, probability bounds, and dense small-system trajectory agreement are tested. The maximum production invariant error, about 2.54e-13, is far below the 1e-9 gate and too small to explain the central-charge uncertainty.

Table 10. Headline statistical precision

Model	SE	95% half-width	|estimate-target|	|difference|/SE
Clean Ising	0.005225	0.010172	0.001261	0.241
Nishimori random-bond Ising	0.008181	0.016120	0.007531	0.921
Weak self-dual Majorana network	0.004240	0.008373	0.002893	0.682

Bootstrap percentile intervals need not be exactly symmetric around the reported mean, so the half-width is descriptive rather than a substitute for the stored endpoints.

Finite-size bias is probed by changing L_min and correction terms. This is not an invitation to choose the fit closest to a target. The primary fit is frozen, and alternatives are diagnostics. For weak self-duality, paired shifts exploit common underlying blocks. For Nishimori, the paired bootstrap interval for the L_min difference must contain zero. For clean Ising, both exact and Monte Carlo windows remain visible.

A 95% confidence interval has a repeated-sampling interpretation under the analysis assumptions. It does not assign a 95% probability that a fixed true value lies inside the realized interval, and it does not include unknown model misspecification. The report therefore says that a target is consistent with the data, not that the target has been proved.

ERROR-BUDGET DISCIPLINE

More samples primarily reduce stochastic uncertainty. Larger widths and better correction models primarily address asymptotic bias. Exact oracles address implementation error. These remedies are not substitutes: a very long run of the wrong model produces a precise wrong answer.

SECTION 09

Implementation and Reproducibility

The code is organized around narrow scientific interfaces. Configuration parsers validate widths, sample budgets, tolerances, and fixed critical constants before output directories are populated. Geometry modules know which degrees of freedom interact. Numerical kernels update a lattice, transfer vector, or covariance matrix. Samplers own stream state and emit block records. Schema modules serialize records. Python loaders reject incompatible artifacts before fitting.

Resumability is stream based. A completed stream artifact includes its schema version, full configuration, stream identity, and estimate blocks. A manifest stores its SHA-256 digest. On restart, compatible artifacts are reused byte for byte; a mismatched digest or configuration stops the run. This is safer than appending to one giant file because interruption cannot leave an ambiguous half-record that later analysis accepts.

ANALYSIS DATA FLOW

```
manifest, blocks, oracles = validate_and_load(run_dir)
summary = aggregate_with_model_specific_covariance(blocks)
fits = fit_finite_size_family(summary)
bootstrap = resample_declared_independent_units(blocks)
gates = evaluate_predeclared_checks(fits, oracles, diagnostics)
write_processed_tables_plots_and_report(summary, fits, gates)
```

The model-specific covariance step is intentionally explicit. A generic row-wise bootstrap would be shorter code but scientifically incorrect for at least one of the three datasets.

Small exact tests have disproportionate value. The clean transfer action can be compared with a dense matrix at tiny L. The Nishimori disorder generator can be checked against configured bond probabilities and a small transfer product. The Majorana covariance update can be compared with dense Hilbert-space evolution across every short Born trajectory. These tests establish signs and normalizations before production-scale statistics make failures difficult to localize.

Production gates are data, not hidden conditions in prose. Each gate stores a name, criterion, observed value, required flag, and pass status. The analysis exits nonzero when a required gate fails, while diagnostic configurations can explicitly mark gates as non-production. The report lists every gate, which prevents a favorable headline result from hiding failed sampling or physics checks.

Table 11. Implementation principles

Principle	Concrete mechanism	Failure prevented
Determinism	Keyed Xoshiro256++ streams	Thread scheduling changes the sample
Atomicity	Temporary file followed by replacement	Partial artifact appears valid
Compatibility	Schema and full-configuration equality	Mixed runs are analyzed together
Integrity	SHA-256 artifact manifest	Silent file corruption or replacement
Separation	Rust sampling; Python analysis	Exploratory plotting contaminates kernels
Predeclaration	Frozen primary fit and required gates	Selecting favorable diagnostics
Independent oracles	Exact identities and dense small systems	Correct-looking output from wrong equations

These principles are implementation choices because each blocks a known path to a scientifically misleading result.

Reproduction begins with the frozen configurations and lock files. The Rust test suite should run before any production command. Python analysis tests should then validate loaders, fits, bootstraps, and report fields. A production rerun is expensive and is not necessary to regenerate this integrated report: its inputs are the already validated processed files and charts, whose hashes appear in the appendix.

LANGUAGE BOUNDARY

Rust owns random draws and numerical state evolution. Python owns data processing and visualization. The integrated report generator performs no Monte Carlo sampling, so rebuilding HTML or PDF cannot change a scientific estimate.

SECTION 10

Open Research: Learning-Induced Metal-Insulator Transition

EXPLORATORY RESULT—NOT A FOURTH BENCHMARK CARD

This chapter is deliberately separated from the three verified central-charge benchmarks. Its frozen status is xy_reproduced_diii_inconclusive. The XY validation gate passed, but the generic DIII scan was inconclusive under the predeclared phase-persistence rule.

The open question is whether changing the physical measurement axis in a monitored surface-code tensor network drives a transition between extended, metal-like Majorana correlations and localized, insulator-like correlations. Unlike the preceding benchmark chapters, no target DIII central charge was supplied. The calculation must first reproduce a known transition on the special XY line and only then search a generic symmetry-class-DIII cut without selecting a favorable point after seeing the data.

```
sigma(theta,phi) = sin(theta) cos(phi) X + sin(theta) sin(phi) Y + cos(theta) Z
```
(25)

The XY validation line fixes theta/pi=0.5. The exploratory generic cut fixes theta/pi=0.45; nonzero polar and azimuthal components remove the special class-D block decomposition.

Rust generated every conditional Born outcome with Xoshiro256++ and evolved a real antisymmetric Gaussian covariance matrix. Rational measurement updates were followed by outcome-dependent orthogonal rotations. An orthogonal polar projection after each period removed floating-point drift in Gamma^2=-I. Python read only frozen block data, compared entanglement-arc models, evaluated phase evidence, and rendered the reports; it performed no Monte Carlo evolution.

PREDECLARED TWO-STAGE DECISION

```
run_xy_validation(theta_pi=0.50)
if xy_bracket overlaps reference_window:
    scan_generic_diii(theta_pi=0.45)
    publish_candidate_only_if_adjacent_phase_evidence_persists
else:
    status = validation_failed
```

The branch structure prevents a generic-DIII claim when the known XY transition is not reproduced. A missing DIII bracket is retained as an inconclusive exploratory result rather than repaired by post hoc scans.

Table 12. Frozen open-research decision

Quantity	Frozen value	Claim class
XY bracket phi/pi	[0.24, 0.25]	validation
XY reference window	[0.20, 0.28]	predeclared validation
DIII bracket	none	exploratory / inconclusive
Casimir c_eff alpha	not fitted	exploratory / not published
anisotropy alpha	unstable	exploratory / not published
DIII central charge	not published	exploratory / not published
runtime	344.982 s	within 3300 s ordinary limit

Every DIII entry is exploratory. Null estimates are scientific outcomes of the claim gates, not missing report fields.

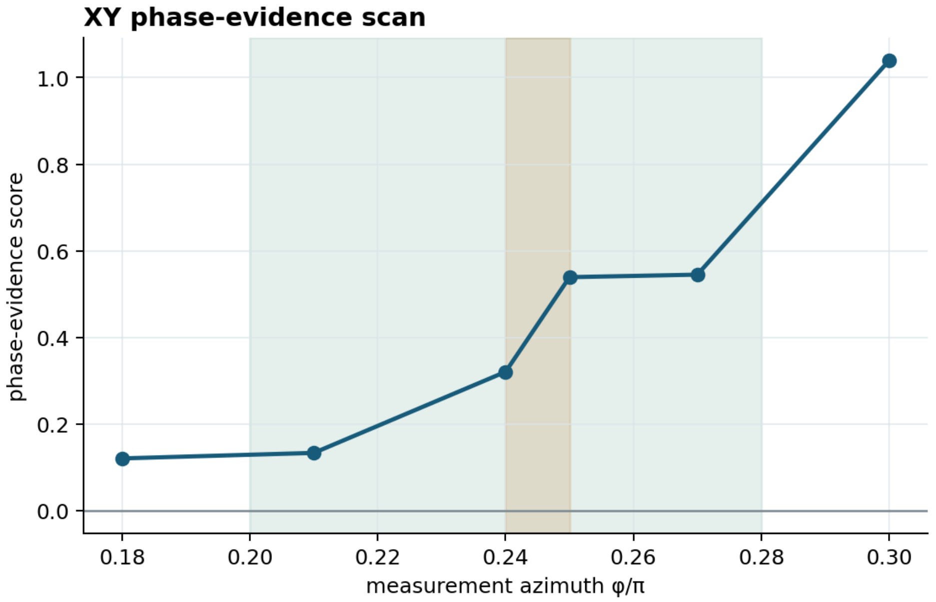

Figure 22. Validation scan: the frozen XY bracket is [0.24, 0.25], inside the declared reference window. *Interpretation limit:* This figure validates the implementation on a special class-D line; it does not establish a generic-DIII transition.

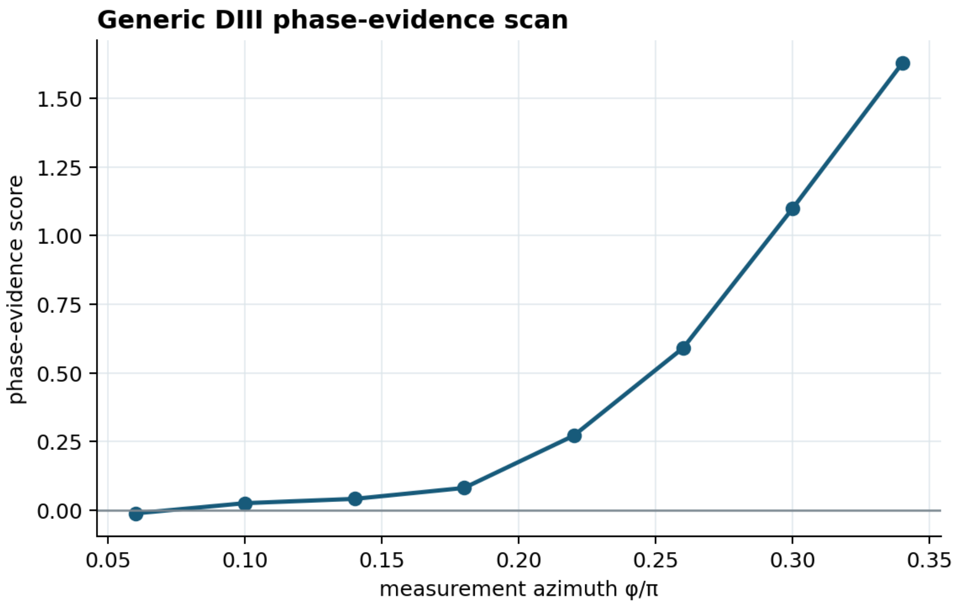

Figure 23. Exploratory generic-DIII scan. Scores evolve smoothly across the grid, but no adjacent pair satisfies the full persistent opposite-phase rule. *Interpretation limit:* The score is a composite finite-size diagnostic, not a central charge or a standalone order parameter with a universal zero.

Table 13. Exploratory DIII evidence scan

phi/pi (exploratory)	evidence score (exploratory)	label
0.06	-0.011280	exploratory
0.10	0.025868	exploratory
0.14	0.041543	exploratory
0.18	0.081359	exploratory
0.22	0.271665	exploratory
0.26	0.591321	exploratory
0.30	1.099846	exploratory
0.34	1.627982	exploratory

All values in this table are exploratory diagnostics. They are not published transition coordinates or universal quantities.

The physical Born and deliberately nonphysical IID-sign controls differ strongly: their frozen means are 2.374261 and 2.525391, with z=21.80. This confirms that unconditional random signs cannot replace state-conditioned Born draws. Scientific oracles passed, all 4 widths (8, 12, 16, 24) and 4 streams per point completed, and no runtime reserve was needed.

WHAT REMAINS UNRESOLVED

The present data do not publish a DIII transition, Casimir amplitude, anisotropy, or central charge. A future study should add independent streams and widths around the low-phi crossover, predeclare a sharper refinement grid, and test whether phase labels persist under alternative finite-size windows. Inconclusive does not mean that a transition is absent.

SECTION 11

Conclusions

The clean Ising calculation recovers the expected c=1/2 through two independent numerical routes. The transfer result 0.499424 demonstrates that the finite-size coefficient and sign conventions are correct, while the Monte Carlo result 0.498739 demonstrates that cluster sampling and thermodynamic integration reproduce the same physics within uncertainty.

The Nishimori calculation verifies the ordinary quenched target 0.464, obtaining 0.456469. The result is not the approximately 0.522 quantity associated with a different replica or Born weighting. Common-disorder width vectors, paired bootstrap analysis, the Nishimori energy identity, and the bond-frequency audit collectively support the ensemble interpretation.

The weak self-dual calculation obtains 0.444107, consistent with 0.447 at a 95% half-width below 0.01. Gaussian covariance oracles, Born trajectory enumeration, self-duality diagnostics, ESS, residual tests, and fit variants all pass. Rao-Blackwellization is central to the achieved precision because it removes conditional outcome noise without changing state evolution.

Across all three models, central-charge verification is best understood as a chain of controlled reductions. Microscopic dynamics produce block observables; blocks produce finite-size estimates with the correct covariance; scaling fits isolate a small universal term; resampling quantifies stochastic variation; alternative fits expose sensitivity; and independent oracles test the chain at points where agreement with a target cannot.

FINAL ASSESSMENT

The frozen evidence supports all three declared benchmarks at its stated resolution. The strongest conclusion is methodological: exact baselines, ensemble-aware resampling, structural oracles, and transparent systematic checks make central-charge estimates auditable rather than merely close.

SECTION 12

Appendices

The appendices collect definitions and audit information so that the main chapters can remain readable. Parameter meanings are given in the model chapters; the glossary standardizes symbols that recur across different normalization conventions. Provenance hashes bind every displayed input to an exact byte sequence.

Table 14. Equation and notation glossary

Symbol	Meaning
L	Cylinder or strip circumference; the finite-size scaling variable
M	Longitudinal size of a torus or strip
K, K_c	Dimensionless coupling and clean critical coupling
p, K_N	Negative-bond probability and Nishimori-line coupling
phi_L	Quenched log-partition density at width L
Gamma	Real antisymmetric Majorana covariance matrix
gamma_1(L)	Weak self-dual Shannon free-energy rate
c	Central charge in the clean unitary benchmark
c_eff	Effective Casimir coefficient in disordered or monitored ensembles
SE	Estimated standard error of an estimator
95% CI	Bootstrap confidence interval with nominal 95% coverage
ESS	Effective sample size after accounting for correlation
a	Leading finite-size correction coefficient
R	Number of replicas or independent streams, according to model

A symbol can denote a model-specific observable only where its chapter defines the convention. In particular, phi_L and gamma_1(L) are not the same microscopic quantity.

Table 15. Selected primary references

Topic	Reference
Finite-size CFT	H. W. J. Bloete, J. L. Cardy, and M. P. Nightingale, Phys. Rev. Lett. 56, 742 (1986).
Finite-size CFT	I. Affleck, Phys. Rev. Lett. 56, 746 (1986).
Ising solution	L. Onsager, Phys. Rev. 65, 117 (1944).
Nishimori line	H. Nishimori, Prog. Theor. Phys. 66, 1169 (1981).
Challenge reference	Open quantum criticality benchmark, arXiv:2502.14034 and Quantum Harness issue #122.

References identify the standard finite-size arguments and model context. All numerical claims in this report are taken from the frozen local artifacts listed below.

Table 16. Frozen-input provenance

Model	Relative path	SHA-256
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/manifest.json	93622b24b1fdc16af219876556fd1fd01bf9a98762d25c93d09ba0e47eca70ee
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/processed/analysis_metadata.json	f41888b5294f9cab8185559e43f41dcc561527ae47b9fecbb83f35c0cb3d8388
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/processed/central_charge_fits.csv	43d216f6f82df73a4077c200a783366209aee4ac5347cdc944705f4bb2d37e01
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/processed/diagnostics.csv	96026b98e7d515d70d0f4b2c9b236e12a2fc2716dea5a45706e3cfa7854996cd
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/processed/free_energies.csv	f966847d214da440eb3098247d4f79600e2be4848c118c14c0031de6a48d6386
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/figures/central_charge_comparison.png	adf84b64d7d95dbe42d645e1e2c55c6a497070bbb58f9e96294781bb26bd51c4
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/figures/energy_vs_k.png	e6f20220603f0e5d3bc0796b7329f74b4a8412c068475321d0a2d4833366f3e9
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/figures/fit_stability.png	a9f24f4a5ae8e59f7315ccdc1f03ecea4b53f91e5563ad510d7a9cae568e4f69
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/figures/free_energy_scaling.png	c08e7feb2caa33064c421b01bf7952d76e167354babf9b0b11cb676e3b20b5ec
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/figures/integration_convergence.png	5032e6f5b2e7a591ca4ce385bc73e1a31f52c1657e7acb0407184283635662ae

Model	Relative path	SHA-256
Clean Ising	tracks/qmc/results/clean-ising-20260729-120302/figures/replica_diagnostics.png	f68405822b2a8bc3c129cfa537ffe2b51a6596a1c48846975ac7cf9d515ed4ab
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/manifest.json	46fea764b5175f73be26d2feccb70f024be2fddf3fdaf60f57244f65e0dfecef
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/processed/summary.json	a6890589b3396acd3d6b1041cf1b78a22a8b400d9bd5058638d73f6fe5f97e74
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/processed/gates.json	b491e05348538d391658e54acc86224aa63d15c8a0dd133af6456cbee4166e7c
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/processed/free_energy.csv	e7d03dadd695502c3a16968890430760cddede5132b4cbc37e75061392bcb16e
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/processed/central_charge_bootstrap.csv	1754592dc95eefe94c6851156c10db20ef4c5d6ada593928aba7722fcd920ed2
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/figures/central_charge_bootstrap.png	39dc60e5892aa4d893d1c7bb9d68541a613d2b5e1b884e122edd9ad5ceec2d41
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/figures/fit_window_stability.png	f1a1a958d129a47d62045a5ad40cba6387fb0b08b5d058f8f78b9e80faec4742
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/figures/free_energy_fit.png	5c57061a37e123bb9006c08a78260091409315f459d355c357e5e3b7ac2684df
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/figures/negative_bond_frequency.png	bde362cc4ee1139a29c3ed331e2e45ae42a05fae7083b5050d27c87e5772933d
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/figures/nishimori_energy_identity.png	47a61a8c02ce437885a96f61635f43b781e7e9b86655eba70bd2320b8694a44d
Nishimori random-bond Ising	tracks/qmc/results/nishimori-ising-20260729-refinement1/figures/sampling_stability.png	400b0223a092d81de103e0bcbcba3a68f612a28af5fd8f1eded431ff89e8af74
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/manifest.json	8a7f41b09fcab50ecaf894bfd206233948350f5c106382b1871a34f318ad4f24
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/processed/summary.json	f485ed37bc94e377c17cb3f5de9320d5b54323ffa84ebc9bdb424a3752628b3f
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/processed/gates.json	cf086cb6ae742fee5fd5218211a2be49387d7ee1b009d4d5f8d4db4127eccd7f
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/processed/finite_size.csv	9af3c42f3005084271ad32b13cab32e0d3c7c2dc98e970825ea784262a06785a
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/processed/fit_variants.csv	cd36d844421fe6463498b141c190f037f41263140833f11e11be7624c8c593f9
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/figures/convergence-ess.png	0552819e86c43ebe85fb64f3456f22cbd0fcacdccfc9da25ea34625c917ee3a1
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/figures/finite-size-scaling.png	98f0332c16c6f167e20560a3ddbed759ebf6c08ac8355918aaf0445b5a92b0be
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/figures/fit-stability.png	d71c6ffe2db6a53047950aa9490d89f2293eeb2236213f8e35896905976c1a79
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/figures/residuals.png	0d508173b56a7a98f36c1f2547cb01fd2c1bb1b450f0ef5dc3fdf4258f55db15
Weak self-dual Majorana network	tracks/qmc/results/weak-self-dual-20260729-154737/figures/self-duality.png	eab403ae1de692cfcce5d7eed378e386d3e0c99867d75889a0204e898124f5ad

Hashes are calculated by the report loader. A changed byte changes the digest and therefore invalidates an otherwise identical-looking report build.

A reproducibility audit should begin by checking these digests, then run the report test suite, and finally inspect both output formats. Re-running the expensive simulations is a separate operation. If it is performed, the new run must receive a new result directory and must not overwrite the frozen evidence used here.